MIL-C-50797(MU)
25 May 1973
SUPERSEDING
LCA-PD-58
7 June 1967

MILITARY SPECIFICATION

CASE, CARTRIDGE, 20MM, M103A1

1. SCOPE

1.1 This specification covers the steel cartridge case for use in the assembly of 20mm cartridges. (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-651	- Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D11075813	- Case, Cartridge, 20mm, M103A1
C7258801	- Classification of Defects for Case, Cartridge, 20mm, M103, M103A1 and M103B1
D7258817	- Cartridge, 20mm, Target Practice, M55A2
D7259191	- Cartridge, 20mm, for M55 Type, Target Practice, Projectile Metal Parts Test

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2.1 (Cont)

DRAWINGS (Cont)

D7258856 - Carton, Interplant Shipment, 20mm, M103 Type Cartridge Cases, Empty, Unprimed
IEL 11075813 - Index of Inspection Equipment Lists for: Case, Cartridge, 20mm, M103A1

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistics Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B117-64 -- Method of Test for Salt Spray (Fog) Testing
ASTM E18-67 -- Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM A317-54 (1965) -- Macroetch Testing and Inspection of Steel Forgings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

3. REQUIREMENTS

3.1 General.-The cartridge case shall comply with the requirements of Drawing (Dwg.) D11075813, referenced specifications and other requirements specified herein.

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3.2 Manufacturing process.-The cartridge case shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.3)

3.3 Head rim thickness, maximum.-The maximum head rim thickness of each cartridge case shall be in accordance with Dwg. D11075813.

3.4 Hardness.-The cartridge case shall comply with the Rockwell hardness pattern specified by Dwg. D11075813.

3.5 Cold shuts.-The cartridge case shall be free of cold shuts.

3.6 Protective coating.

3.6.1 Cleaning.-The cartridge case shall be clean of all foreign matter prior to plating. The cleaning process shall be one which is suitable for use with the process for zinc coating, electrodeposited.

3.6.2 Plating.-The completely formed cartridge case shall be coated complying with applicable requirements of Dwg. D11075813.

3.6.3 Corrosion resistance.-The plated exterior of the cartridge case shall show no visible signs of white corrosion products on exterior surfaces, including accidental scratches through the chromate film to the zinc plate, after 48 hours exposure to the salt spray.

3.7 Function and casualty.-The sample cartridge cases shall be assembled into cartridges and shall function without casualty when tested as specified in Table I. (see 4.3.4)

3.8 Workmanship.-The requirements for workmanship are as specified by the applicable drawings, referenced specifications and the following:

3.8.1 Processing defects.-The cartridge case shall be free of visually defective metal, folds, wrinkles, bulges, buckles, dents, draw scratches, damaged plating, splits, perforations and other forms of substandard workmanship.

3.8.2 Cleanliness.-The cartridge case shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government

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4.1 (Cont)

reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 5100 cases. The sample shall be manufactured using the same equipment, processes and procedures as will be used in regular production. Sample shall be divided and tested as follows:

100 Cases	-- Physical and dimensional examination and test
2500 Cases	-- Function and casualty test, ambient temperature, M61 gun
500 Cases	-- Function and casualty test, ambient temperature, M39 gun
500 Cases	-- Function and casualty test, -65°F, M61 gun
500 Cases	-- Function and casualty test, -65°F, M39 gun
500 Cases	-- Function and casualty test, +165°F, M61 gun
500 Cases	-- Function and casualty test, +165°F, M39 gun

The acceptance criteria specified in Table II for the combined sample will be used and applied independently to each of the above phases of the function and casualty test. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. Identification shall be in accordance with MIL-STD-1168.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified by the contract. (see 6.4)

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications will result in sample disapproval.

4.3 Quality conformance inspection.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

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4.3.1.1 Lot.-A lot shall consist of cartridge cases produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions and same specification and specification revision. Not more than one heat of steel shall be used in one case lot except that one case lot may contain cases processed from the subsequent heat of steel and the records shall be so annotated.

4.3.1.2 Lot identification.-Each cartridge case and each packed cartridge case lot shall be identified in accordance with applicable drawings and MIL-STD-1168.

4.3.2 Examination.-Each cartridge case shall be examined for head rim thickness, maximum, requirement of 3.3 in accordance with the method as specified in 4.4.5. Examination for major and minor defects shall be performed as specified herein. Each cartridge case found to be defective shall be removed from the lot.

4.3.2.1 Sampling and acceptance criteria.-The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.3.2.2.c) In addition, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.5)

4.3.2.2 AQL's.-The following AQL's are assigned to major and minor defects.

a. Individual defects.

Major defects	0.25%
Minor defects	0.40%

b. Class basis.

Major defects	1.50%
Minor defects	2.50%

c. Individual defect characteristic associated with the class basis.

Major defects	0.40%

4.3.3 Classification of defects.-The classification of defects shall be as follows:

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4.3.3.1 Case, Cartridge (see Dwg. D11075813) 1/

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Metal defective	Visual
102. Flash hole missing or obstructed	Visual
103. Corrosion	Visual
104. Oil or grease	Visual
105. Head configuration	Visual
106. Thickness, head rim, min.	Gage
107. Perpendicularity case head with datum surface	Gage
108. Chamfer, head rim, missing	Visual
109. Length of flat on head rim, min.	Gage
110. Diameter of head, min.	Gage
111. Diameter, primer pocket	Gage
112. Depth, primer pocket	Gage
113. Runout, primer pocket with datum surface	Gage
114. Diameter, extractor groove, max.	Gage
115. Width, extractor groove, min.	Gage
116. Wall thickness, .375 inch from inside base	Gage
117. Wall thickness, .800 inch from inside base	Gage
118. Wall thickness, 2.664 inch from inside base	Gage
119. Inside diameter, mouth, min.	Gage
120. Profile and alignment, max.	Gage
121. Length to .942 basic diameter	Gage
122. Total length, max.	Gage
123. Diameter, top of body taper, min.	Gage
124. Diameter, bottom of body taper, min.	Gage
125. Protective finish, missing or incomplete	Visual
126. Split or perforated case	Visual
127. Draw scratch	Visual
128. Convexity or concavity of head surface	Gage
Minor:	
201. Foreign matter other than oil, grease or corrosion	Visual
202. Scratch (other than draw scratch)	Visual
203. Fold, wrinkle or buckle	Visual
204. Radius at mouth of primer pocket missing	Visual
205. Wall thickness, neck area	Gage
206. Total length, min.	Gage
207. Diameter, extractor groove, min.	Gage
208. Width, extractor groove, max.	Gage
209. Thickness, head surface to inside base, min.	Gage

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4.3.3.1 (Cont)

Categories and Defects	Method of Inspection
Minor: (Cont)	
210. Diameter, flash hole	Gage
211. Diameter, head, max.	Gage
212. Inside diameter, mouth, max.	Gage
213. Weight	Balance
214. Marking incorrect, incomplete or unidentifiable	Visual
215. Dent	Visual

1/ MIL-STD-651 shall apply in defining and evaluating visual defects, as applicable to M103A1 case.

4.3.4 Tests.-The tests listed in Table I shall be performed on each case lot in accordance with the test methods and procedures prescribed by 4.4. Sample size and acceptance criteria for each test shall be as specified. Only cartridge cases having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cases shall be thoroughly mixed before being divided into samples for the various tests.

Table I

Test		Sample Size	Reqt Para
Hardness	1/	13	3.4
Cold shut	2/	13	3.5
Corrosion resistance	3/	13	3.6.3
Function and casualty	4/		
Excess pressure			
Ambient temperature		50	
Service pressure			
Ambient temperature		400	
Low temperature (-65°F)		50	
High temperature (+165°F)		50	

1/ Failure of two or more sample cartridge cases to comply with the average hardness requirement at one or more locations shall be cause for rejection of the lot. If one cartridge case of the sample fails to comply with the requirements, a second sample consisting of the same number of cartridge cases as the first sample shall be tested. If in the accumulated sample, two or more cartridge cases fail to comply with the average hardness requirements, the lot shall be rejected.

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4.3.4 (Cont)

2/ The lot shall be rejected if one or more units of the sample shows a cold shut extending beyond the area identified by tangents AB and CD of Figure 1.

3/ Failure of two or more units of the sample to comply with the corrosion resistance requirement shall be cause for rejection of the lot. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

4/ The lot will be rejected if any malfunction or firing casualty of Table II occurs in number(s) equal to or greater than the applicable rejection (Rej) number. If malfunctions or casualties occur in excess of the applicable acceptance (Acc) number, but less than the "Rej" number, a second sample of double the number of the first will be divided and fired in equal numbers in the M39 and M61 guns. The lot will be rejected if in the combined samples malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number. The excess pressure, ambient, high and low temperature phases will be considered as separate tests.

Table II

Function and Casualty Test

Case Casualty A/	Excess Pressure Acc	Excess Pressure Rej	Ambient Acc	Ambient Rej	Low Temp. Acc	Low Temp. Rej	High Temp. Acc	High Temp. Rej
Failure to chamber	0	1	0	1	0	1	0	1
Failure to extract	0	1	0	1	0	1	0	1
Blown primer	0	1	0	1	0	1	0	1
Split B/								
I or S	1	4	6	17	1	4	1	4
J	0	3	2	7	0	3	0	3
K, L or M	0	1	0	1	0	1	0	1
Rupture, partial B/								
S or J	0	3	2	7	0	3	0	3
K or L	0	1	0	1	0	1	0	1
Rupture, circumferential separation	0	1	0	1	0	1	0	1
Detached metal								
Rim shear	0	1	0	1	0	1	0	1
Neck shear C/	0	3	2	7	0	3	0	3

A/ For defect definitions, see AMCR 715-505, Volume 8. The case lot will not be penalized for the occurrence of malfunction(s) which are determined to have been caused by the test weapon or associated equipment.

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4.3.4 (Cont)

B/ Refer to Dwg. C7258801 for classifying splits and ruptures. If a split or partial rupture extends into two or more defined areas, only the most severe defect criterion of Table II will be used.

C/ A case from which a section or a ring segment is missing from the neck area. The lot will not be penalized for the detachment of metal shavings from the crimped area of the case interior.

4.3.4.1 Unlisted firing defects.-The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification, occurring in any firing test, indicates that the product is unsuited for the purpose intended.

4.3.5 Packing and marking inspection.-Inspection for packing and marking shall comply with applicable requirements of MIL-STD-1169.

4.3.6 Inspection equipment.-Index of Inspection Equipment List No. IEL 11075813 identifies the applicable Inspection Equipment List required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding, Dwg. B11057228, a detail of IEL 11075813).

4.4 Test methods and procedures.

4.4.1 Hardness.-The method of test shall be as specified in ASTM E-18-67. The Rockwell scales referenced for use in testing the case hardness at the various locations, shall be as specified by Dwg. D11075813. Each unit of the test sample shall be sectioned as shown by that drawing. The head sections shall have approximately parallel surfaces finished to 63 micro-inches or finer, care being taken not to change the hardness in obtaining that finish. Hardness determination shall be made at the body and neck locations as shown on Dwg. D11075813, Section "A" on both halves of the sectioned case. The four hardness values from locations the same distance from the case head shall be averaged. Hardness determinations shall be made on one surface of head section at the locations as shown on Dwg. D11075813, Section "B." The two or four hardness values, as applicable, that are the same distance from the case head shall be averaged. The average hardness values shall be used to determine compliance with requirements.

4.4.2 Cold shut.-The method of test shall be as specified in ASTM A317-54. The specimens for test shall be the head sections from the sample prepared for determination of the hardness requirements of 4.4.1. Macroscopic examination of the etched test specimens for cold shuts within the area identified by Figure 1 shall be made under 75-100 magnifications.

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4.4.3 Corrosion resistance.-The method of test shall be as specified in ASTM B117-64. The test sample should be air dried for 24 hours prior to testing, to permit sufficient time for the chromate film to harden. The finish shall not be disturbed. The sample shall not be penalized for the presence of white corrosion products on corners or edges.

4.4.4 Function and casualty.-The method of test shall be as specified in AMCR 715-505, Volume 8. The test will be performed at the facility designated by the procuring activity. The test weapon barrel(s) will be at ambient temperature for the firing of each burst. All expended sample cases will be examined for the defects of Table II.

4.4.4.1 Excess pressure.-The sample test cases will be loaded and assembled into cartridges as specified by Dwg. D7259191. The test cartridges will be conditioned for not less than 2 hours at a temperature between 68°F and 72°F prior to firing and while at that temperature, the test cartridges will be fired in 25 shot bursts, divided equally between Guns, 20mm, M39 and M61.

4.4.4.2 Service pressure.-The sample test cases will be loaded and assembled into cartridges as specified by Dwg. D7258817. These test cartridges will be used in the tests specified in 4.4.4.2.1, 4.4.4.2.2 and 4.4.4.2.3.

4.4.4.2.1 Ambient temperature.-The test cartridges will be brought to ambient temperature and fired in 25 shot bursts, divided equally between Guns, 20mm, M39 and M61.

4.4.4.2.2 Low temperature.-The test cartridges will be conditioned for not less than 6 hours at a temperature between -65°F and -70°F and while at that temperature, the test cartridges will be fired in 25 shot bursts, divided equally between Guns, 20mm, M39 and M61.

4.4.4.2.3 High temperature.-The test cartridges will be conditioned for not less than 6 hours at a temperature between +165°F and +170°F and while at that temperature, the test cartridges will be fired in 25 shot bursts, divided equally between Guns, 20mm, M39 and M61.

4.4.5 Head rim thickness, maximum.-The method of examination for head rim thickness, maximum, shall be approved by the contracting officer. The equipment design shall provide control(s) that will assure rejection of defective cartridge cases.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.-Each lot of cartridge cases shall be packed as specified by Dwg. D7258856 and in accordance with note for Case, Cartridge, M103B1.

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5.2 Marking and labeling.-Marking or labeling of cartons shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use.-These cartridge cases are intended for ammunition to be used in 20mm automatic gun systems that have been designed for firing cartridges having the M103 case configuration.

6.2 Ordering data.-Invitation for bids and contracts or orders will specify the following:

6.2.1 Title, number and date of this specification.

6.2.2 Place of inspection, if not at place of manufacture.

6.2.3 Provision for contractor submission of a manufacturing process and head rim thickness examination for approval. (see 3.2 and 3.3)

6.2.4 Packing and marking instructions. (see 5)

6.2.5 Provisions for shipment of first article and regular production lot samples for function and casualty test.

6.2.6 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.3 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item of contract.

6.4 First article sample.-In order to minimize inspection costs, the firing tests of 4.3.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cases may be required by the testing facility.

6.5 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

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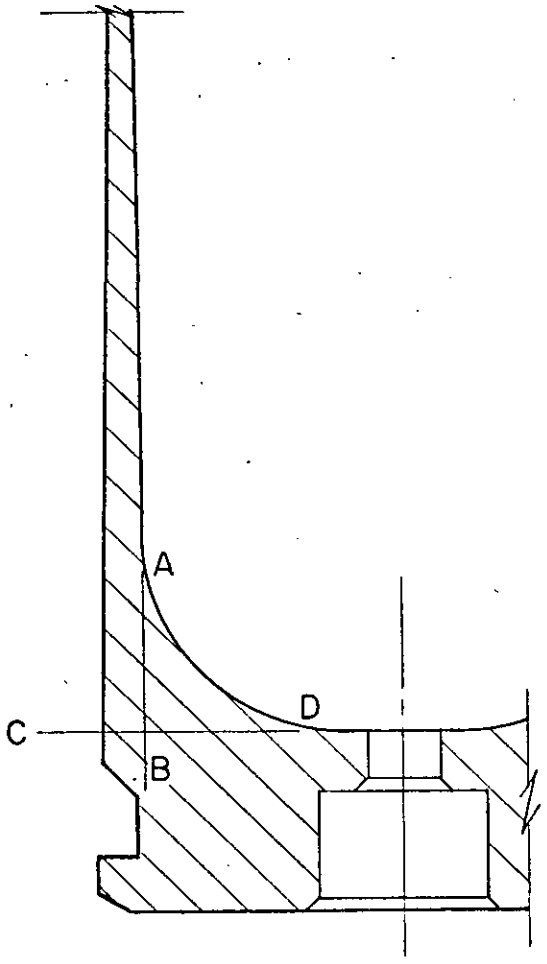
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TANGENTS AB AND CD INDICATE LIMITS BEYOND WHICH FOLDS OR COLD SHUTS MUST NOT EXTEND IN ACCEPTABLE CASES.

FIGURE 1

(CASE, CARTRIDGE, 20MM, M103A1)

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-50797(MU) Case, Cartridge, 20MM, M103A1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B., RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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